

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final
Duration: 2 hours

Semester: Fall 2024
Full Marks: 50

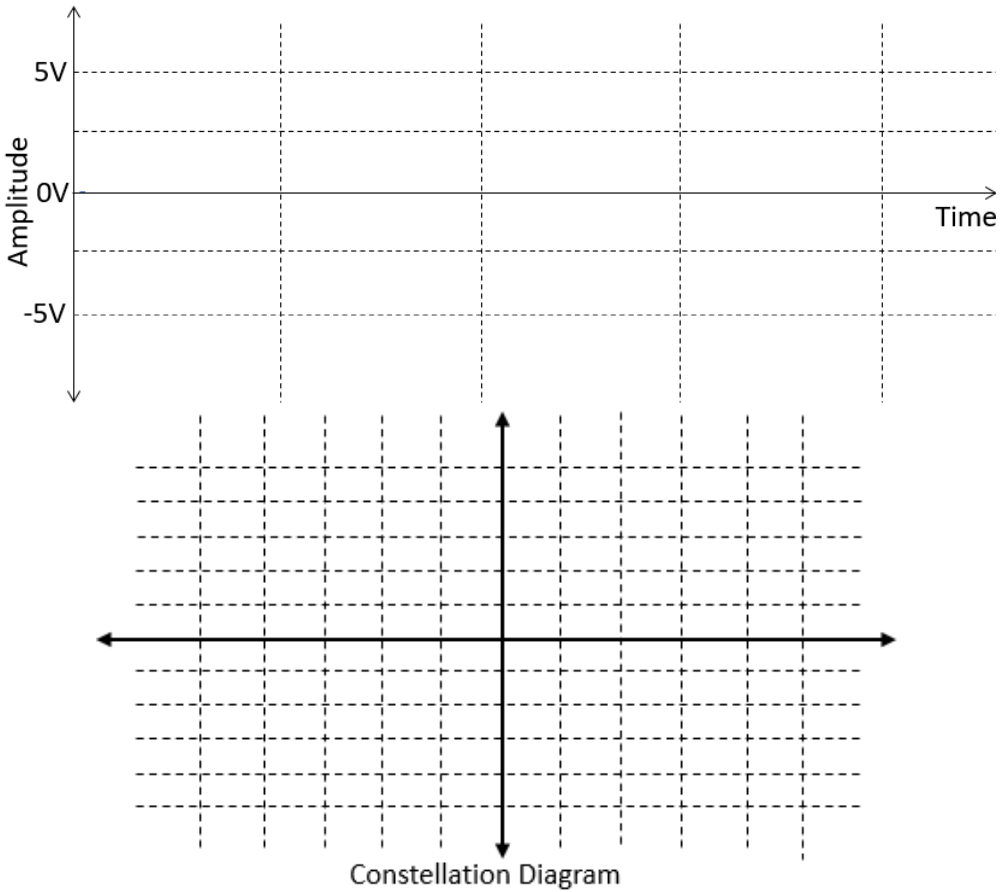
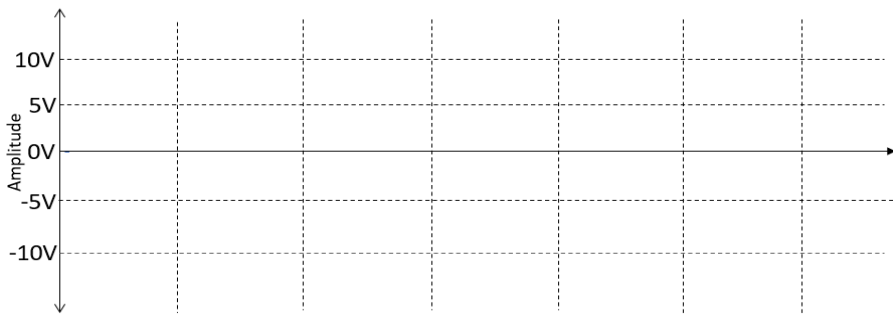
CSE 320/EEE361/ECE361: Data Communications

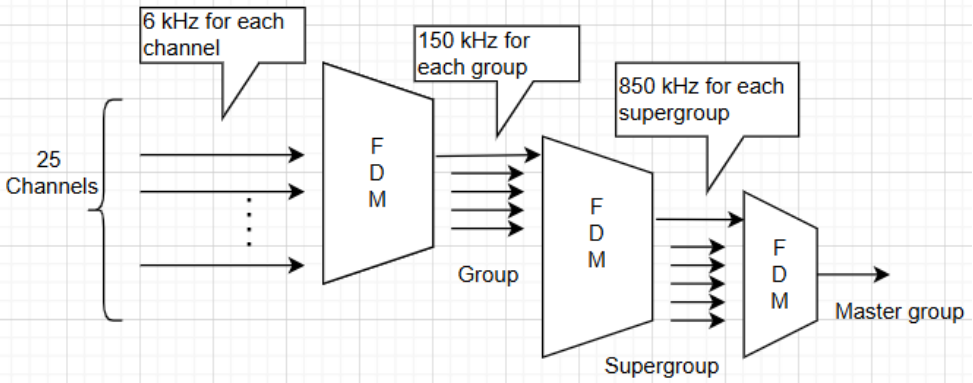
Answer the following questions.
Figures in the right margin indicate marks.

SET A

Name:	ID:	Section:
-------	-----	----------

1. [CO2]	a) The following table depicts a sampled analog signal for digital signal representation. By applying the concept of Pulse Code Modulation, assume there will be 4-bit code words for each sampled amplitude. Show the normalized quantized value and encoded word for the given analog signal value at different time stamps. Assume that the sampling amplitudes are between -25V to +63V. <table><tr><th>Time</th><th>Analog Signal Value (V)</th></tr><tr><td>0</td><td>23.3</td></tr><tr><td>1</td><td>-12.7</td></tr><tr><td>2</td><td>46.8</td></tr><tr><td>3</td><td>1.4</td></tr></table>	Time	Analog Signal Value (V)	0	23.3	1	-12.7	2	46.8	3	1.4	[6]
Time	Analog Signal Value (V)											
0	23.3											
1	-12.7											
2	46.8											
3	1.4											
2. [CO2]	a) A QPSK (Quadrature Phase Shift Keying) system transmits 2 bits per signal element with the following information: <table><tr><th>Bit Pair</th><th>Coordinates (I, Q)</th></tr><tr><td>00</td><td>(2,2)</td></tr><tr><td>01</td><td>(2, -2)</td></tr><tr><td>10</td><td>(-2,2)</td></tr><tr><td>11</td><td>(-2, -2)</td></tr></table> The carrier signal contains 2 cycles for each signal element, and each cycle lasts 1 ms. Given the bit sequence 10110001, Draw the QPSK waveform and the constellation diagram for an 8 ms transmission. Clearly indicate the phase and amplitude of the modulated waveform.	Bit Pair	Coordinates (I, Q)	00	(2,2)	01	(2, -2)	10	(-2,2)	11	(-2, -2)	[4]
Bit Pair	Coordinates (I, Q)											
00	(2,2)											
01	(2, -2)											
10	(-2,2)											
11	(-2, -2)											

		 Constellation Diagram	
	b)	During another transmission, the phase sequence of a received signal was 315°, 135°, 225°, and 45° respectively. Using the above table, decode these phases into their corresponding bit pairs and find the resulting bit sequence.	[2]
	c)	Design a FSK (Frequency Shift Keying) system to transmit the binary data stream 101101 under the following conditions: Each signal element has a carrier frequency $f_c = 2$ Hz and a frequency shift $\Delta f = 1$ Hz. The peak amplitude is 5V, the phase is 180°, and the duration of each signal element is 1s. • Calculate the frequencies f_1(for binary 1) and f_0 (for binary 0). • Draw the FSK modulated signal for the given binary data stream 101101. Make sure to indicate the frequency changes for each signal element. 	[1+3]

3. [CO3]	a) Suppose 5 data sources with data rates of 140 Kbps, 280 Kbps, 140 Kbps, 560 Kbps, and 280 Kbps are to be multiplexed. Show the visual representation of MUX input and output without adding extra pulses in the input channels and the number of input channels will remain the same. If a character at a time is interleaved using the concept of TDM with 1 synchronizing bit, then answer the following questions: • What is the frame duration? • What is the input bit duration? • What is the frame rate? • What is the output data rate? • What is the output bit duration?	[2+5]
	b) Analog Hierarchy:  In the given analog hierarchy, 25 channels are multiplexed using Frequency Division Multiplexing (FDM), where each channel occupies a bandwidth of 6 kHz, forming a group with a bandwidth of 150 kHz. Five groups are further multiplexed to form a supergroup with a bandwidth of 850 kHz. Finally, six supergroups are combined to form a master group. • Calculate the bandwidth allocated to each guard band before forming the supergroup. • Calculate the total bandwidth of the master group if the guard band between each supergroup is 180 kHz	[4]
	c) What problems might occur if the interleaving order is disrupted during transmission using TDM?	[3]

